a straight line at a uniform speed chosen so that we arrive at B exactly 100 seconds later. If we don't go along a straight line it takes more speed, and our watch is slowed down." But wait! That was before we took gravity into account. Isn't it better to curve upward a little bit and then come down? Then during part of the time we are higher up and our watch will run a little faster? It is, indeed. If you solve the mathematical problem of adjusting the curve of the motion so that the elapsed time of the moving watch is the most it can possibly be, you will find that the motion is a parabola—the same curve followed by something that moves on a free ballistic path in the gravitational field, as in Figure 6-19. Therefore the law of motion in a gravitational field can also be stated: *An object always moves from one place to another so that a clock carried on it gives a longer time than it would on any other possible trajectory*—with, of course, the same starting and finishing conditions. The time measured by a moving clock is often called its "proper time." In free fall, the trajectory makes the proper time of an object a maximum.

Let's see how this all works out. We begin with Eq. (6.5) which says that the *excess* rate of the moving watch is

$$\frac{\omega_0 g H}{c^2}. \tag{6.13}$$

Besides this, we have to remember that there is a correction of the opposite sign for the speed. For this effect we know that

$$\omega = \omega_0 \sqrt{1 - v^2/c^2}\,.$$

Although the principle is valid for any speed, we take an example in which the speeds are always much less than c. Then we can write this equation as

$$\omega = \omega_0(1 - v^2/2c^2),$$

and the defect in the rate of our clock is

$$-\omega_0 \frac{v^2}{2c^2}. \tag{6.14}$$